

Advanced Machine Learning (Semester 2 2026)

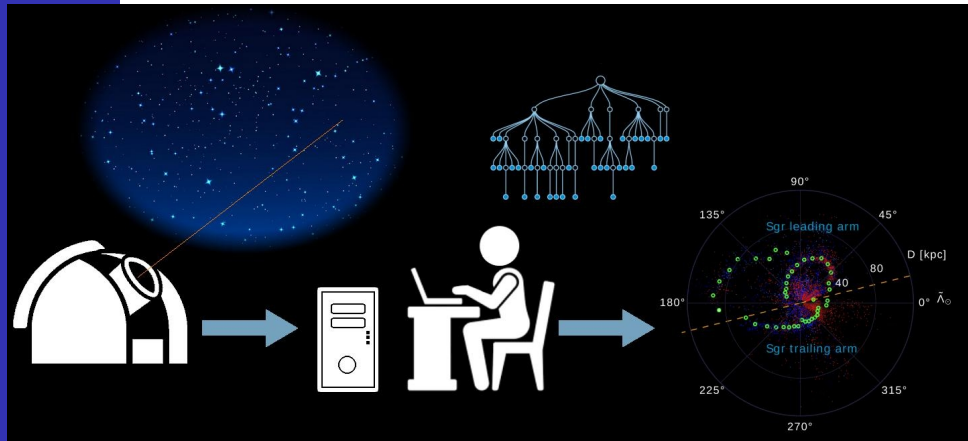
Introduction & Course Logistics

Nina Hernitschek

Centro de Astronomía CITEVA
Universidad de Antofagasta

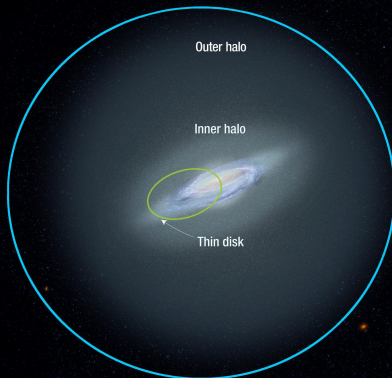
September 28, 2026

Motivation



~ 400 kpc LSST

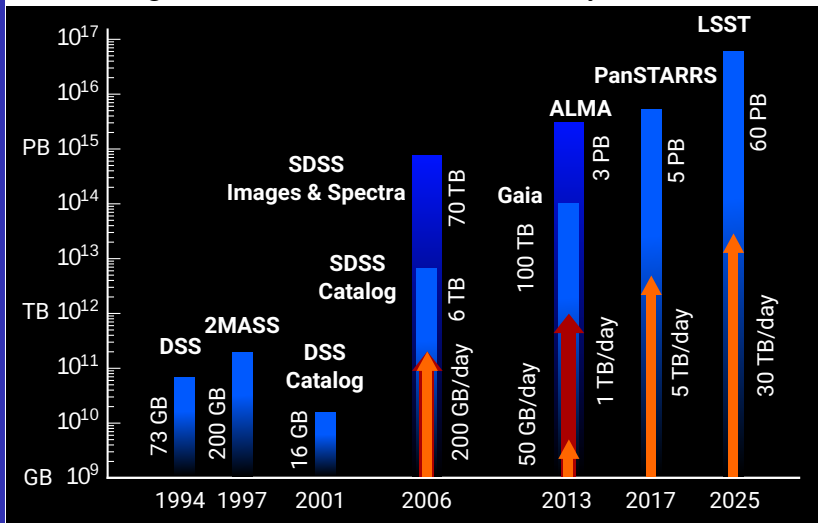
~ 120 kpc PS1 3σ



~ 10 kpc limit of SDSS studies
for kinematics & $[\text{Fe}/\text{H}]$

Challenges in Data Handling

increasing data volume in astronomical surveys



Motivation

Overview

Course

Logistics

Machine

Learning in

Astronomy

Machine

Learning -

Terminology

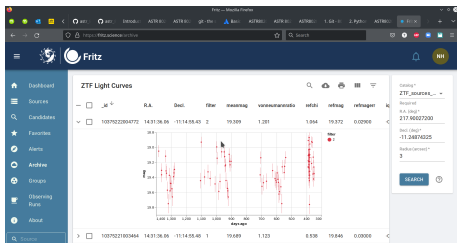
Software

Outlook

what you will learn in this class

this course will prepare you for “doing science” with current and upcoming large astronomical surveys:

science with astronomical data



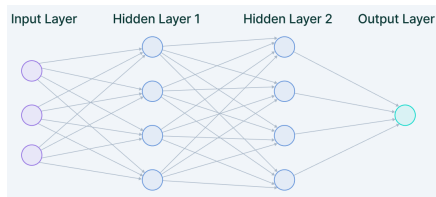
what you will learn in this class

this course will prepare you for “doing science” with current and upcoming large astronomical surveys:

science with astronomical data



artificial neural networks



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

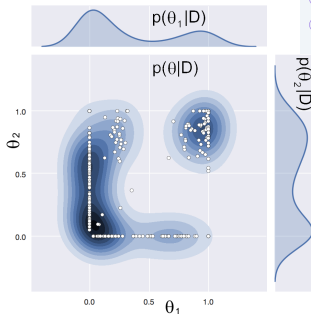
what you will learn in this class

this course will prepare you for “doing science” with current and upcoming large astronomical surveys:

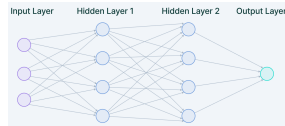
science with astronomical data



statistical methods



artificial neural networks



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

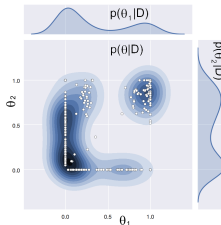
what you will learn in this class

this course will prepare you for “doing science” with current and upcoming large astronomical surveys:

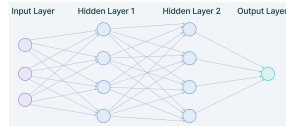
science with astronomical data



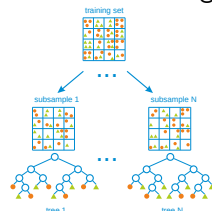
statistical methods



artificial neural networks



machine learning



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

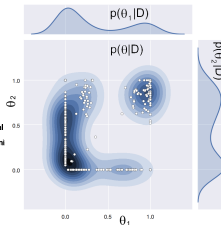
what you will learn in this class

this course will prepare you for “doing science” with current and upcoming large astronomical surveys:

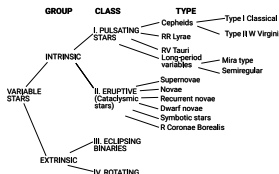
science with astronomical data



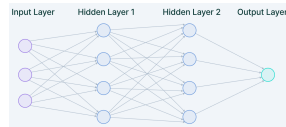
statistical methods



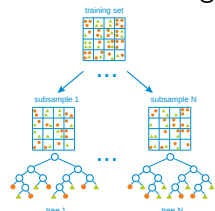
classification



artificial neural networks



machine learning



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Course Logistics

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

course content:

- lecture: Monday 9:00 - 11:00
- practice: Friday 9:00 - 11:00
- preparation of a paper presentation (of your choice)
- identification of a problem related to your research to be solved with machine learning, i.e. neural networks: 3 presentations, report

grading:

- participation: 10 %
- paper presentation: 20 %
- project presentations (project idea + project status + final): 35 %
- project report: 35 %

deliverables: your github repository

contact and course material:

- e-mail: nina.hernitschek@quantof.cl
- github: https://github.com/ninahernitschek/advanced_machine_learning_2026_2

Course Logistics

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

September 28 Lecture 1: Introduction & Course Logistics

October 5 Lecture 2: Artificial Neural Networks (I), **October 9** Practice 1

October 19 Lecture 3: Artificial Neural Networks (II), **October 23** Practice 2 & *presentation on paper*

October 26 Lecture 4: Training of Neural Networks, **October 30** Practice 3

November 2 Lecture 5: Convolutional Neural Networks, **November 6** Practice 4 & *presentation on project idea*

November 9 Lecture 6: Recurrent Neural Networks

November 20 Practice 5

November 23 Lecture 7: Autoencoders, **November 27** Practice 6

November 30 Lecture 8: Reinforcement Learning, **December 4** Practice 7 & *presentation on project status*

December 14 Lecture 9: Architecture of Machine Learning Projects

December 18 *project presentation*

Rules for Coding, Presentations, Report

same as for Advanced Astroinformatics project:

coding: If you have a question when something doesn't work, summarize what you tried - often this will even lead to the solution.

project report and presentation:

- $\text{\LaTeX}$
- figures: all own figures should be in vectorized pdf format
- abstract: concise summary of your project that gives the big picture
- data and aim of project: data description (incl. citation)
- own work: properly cite what is not your own work; discuss how the previous work is similar to or different from your own work
- implementation: medium-level implementation description with libraries/ software frameworks (incl. citation), project milestones $\Rightarrow$ more details than in a research paper
- related work: include both work aimed at similar problems and work that employs similar solutions to yours
- discussion: reflect your approach (strengths, weaknesses, limitations), lessons learned
- bibliography: bibtex/ref mechanism, ADS/Bibtex information

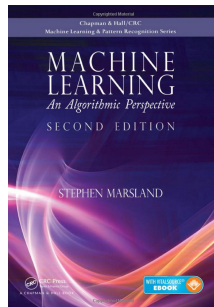
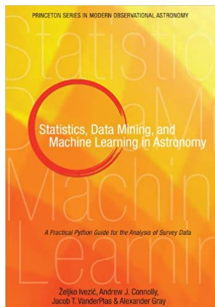
Textbooks

Statistics, Data Mining and Machine Learning in Astronomy: A Practical Python Guide for the Analysis of Survey Data

Ž. Ivezić, A. J. Connolly, J. T. VanderPlas, A. Gray

Machine Learning - An Algorithmic Perspective

Stephen Marsland

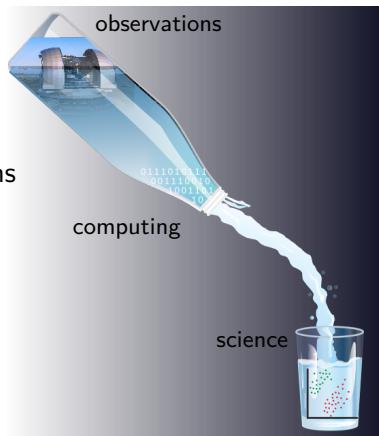


Challenges in Data Handling

astronomy is largely determined by computational capacity

⇒ telescopes & instruments as
front-ends for data processing systems

⇒ **challenge and chance:**
understanding complex phenomena
requires complex data



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

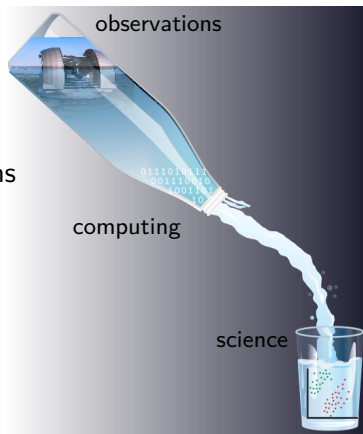
Outlook

Challenges in Data Handling

astronomy is largely determined by computational capacity

⇒ telescopes & instruments as
front-ends for data processing systems

⇒ **challenge and chance:**
understanding complex phenomena
requires complex data



Big Data is transforming how and which discoveries are made

Big Data

Laney et al. 2001: data growth challenge is **three-dimensional**

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Big Data

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Laney et al. 2001: data growth challenge is **three-dimensional**

Big Data is data with at least one big dimension:

- volume
- velocity: bandwidth, response speed
- variety: number and size of individual assets

Big Data

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Laney et al. 2001: data growth challenge is **three-dimensional**

Big Data is data with at least one big dimension:

- volume
- velocity: bandwidth, response speed
- variety: number and size of individual assets

shifting use cases:

As data becomes big data, finding the *right* data has become more important.

Big Data

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Laney et al. 2001: data growth challenge is **three-dimensional**

Big Data is data with at least one big dimension:

- volume
- velocity: bandwidth, response speed
- variety: number and size of individual assets

shifting use cases:

As data becomes big data, finding the *right* data has become more important.

⇒ powerful astrostatistical & machine-learning tools are needed to derive scientific insights

Big Data

shifting use cases:

As data become more plentiful, finding the *right* data has become more important.

⇒ powerful astrostatistical & machine-learning tools are needed to derive scientific insights

Individual measurements giving way to **statistics, clustering, patterns** in the data.

Data processing needs to be **highly automatized**.
Analysis growing more exploratory rather than pre-defined/scripted.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

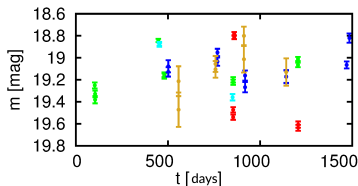
Software

Outlook

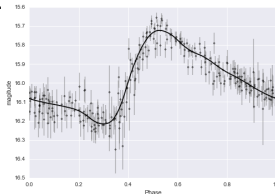
Big Data

one **example** for finding the *right* data :

Pan-STARRS1 3π survey with about 10^9 light curves like that:



goal: finding RR Lyrae* stars whose light curves look like that (if better sampled):



*less than 1 % of the light curves are expected to be from that type

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Statistical Data Analysis

Data-driven methods like statistical methods can reliably **quantify information** embedded in scientific data **without the biases of physical models**.

Requirements:

- find the right method(s): modern statistics is vast in its scope and methodology
- scientific inferences should not depend on arbitrary choices in methodology and variable scale
- correct interpretation of the meaning of a statistical result w.r.t. the scientific goal: (astro-)statistics and machine learning are only tools!

(Astro-)Statistics

a lot is possible:

galaxy clustering

galaxy morphology

weak lensing morphology

strong lensing
morphology

faint source detection

variable source

preclassification



spatial point processes,
clustering

regression, mixture models

geostatistics, density
estimation

shape statistics

false discovery rate

structure functions +
classifier

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

(Astro-)Statistics

a lot is possible:

galaxy clustering

galaxy morphology

weak lensing morphology

strong lensing
morphology

faint source detection

variable source

preclassification

spatial point processes,
clustering

regression, mixture models

geostatistics, density
estimation

shape statistics

false discovery rate

structure functions +
classifier



⇒ **fitting models**

Motivation

Overview

Course
Logistics

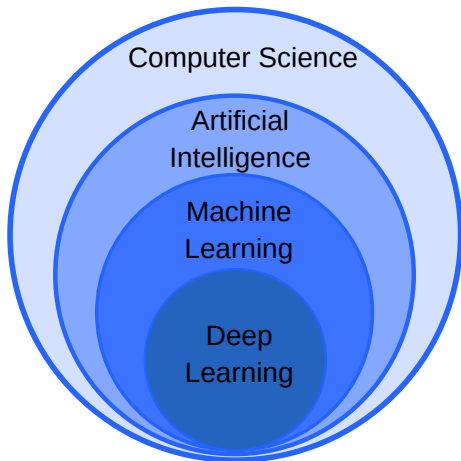
Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Machine Learning - Terminology



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Machine Learning

... is the sub-field of computer science that gives computers the ability to learn without being explicitly programmed
(Arthur Samuel, 1959)

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

**Machine
Learning -
Terminology**

Software

Outlook

Machine Learning

... is the sub-field of computer science that gives computers the ability to learn without being explicitly programmed
(Arthur Samuel, 1959)

⇒ allows to **uncover hidden correlation patterns** through iterative learning by sample data

Motivation

Overview

Course

Logistics

Machine

Learning in

Astronomy

Machine

Learning -

Terminology

Software

Outlook

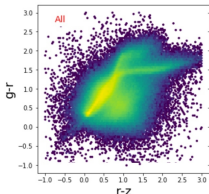
Machine Learning

... is the sub-field of computer science that gives computers the ability to learn without being explicitly programmed
(Arthur Samuel, 1959)

⇒ allows to **uncover hidden correlation patterns** through iterative learning by sample data

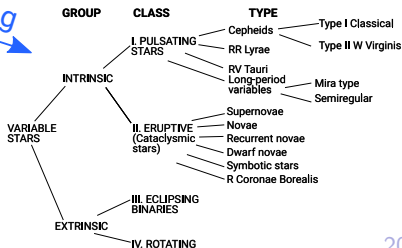
⇒ **in astronomy:**

parameter space of measurements



parameter space of astrophysical objects

machine learning



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Machine Learning

... is the sub-field of computer science that gives computers the ability to learn without being explicitly programmed
(Arthur Samuel, 1959)

⇒ allows to **uncover hidden correlation patterns** through iterative learning by sample data

⇒ **in astronomy**: allows **to model a survey**:

- describing data quality → outlier
- describing light curve characteristics → “features”
- classifying sources → catalogs
- finding substructure → clumps, overdensities, ...

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Functions of Machine Learning Systems

Descriptive

the system uses the data to explain data properties; tools: simple statistical tools such as averages, percentages

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Functions of Machine Learning Systems

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Descriptive

the system uses the data to explain data properties; tools: simple statistical tools such as averages, percentages

Predictive

focuses on predicting and understanding future behavior

Functions of Machine Learning Systems

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Descriptive

the system uses the data to explain data properties; tools: simple statistical tools such as averages, percentages

Predictive

focuses on predicting and understanding future behavior

Prescriptive

the system uses data to make suggestions about actions to take based on the insights gained

Types of Machine Learning Algorithms

there are different types of machine learning algorithms that differ mostly by **how they use data**

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Supervised Machine Learning

labeled data (training data, training set) enable the supervised machine learning algorithm to understand the connection between **features** and **labels**

new observations (target data) are assigned to a group or class



← spiral galaxy



← elliptical galaxy



applications: classification problems, regression problems

Supervised Machine Learning

The objective of a supervised learning model is to predict the correct label for newly presented input data.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

**Machine
Learning -
Terminology**

Software

Outlook

Supervised Machine Learning

The objective of a supervised learning model is to predict the correct label for newly presented input data.

When training a supervised learning algorithm, the **training set** will consist of inputs paired with the correct outputs. Inputs in the training set should represent the **target set** which we have to classify: composition of the data, data quality.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Supervised Machine Learning

The objective of a supervised learning model is to predict the correct label for newly presented input data.

When training a supervised learning algorithm, the **training set** will consist of inputs paired with the correct outputs. Inputs in the training set should represent the **target set** which we have to classify: composition of the data, data quality.

During **training**, the algorithm will search for patterns in the data that correlate with the desired outputs.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Supervised Machine Learning

The objective of a supervised learning model is to predict the correct label for newly presented input data.

When training a supervised learning algorithm, the **training set** will consist of inputs paired with the correct outputs. Inputs in the training set should represent the **target set** which we have to classify: composition of the data, data quality.

During **training**, the algorithm will search for patterns in the data that correlate with the desired outputs.

After training, a supervised learning algorithm will take in new unseen inputs and will determine which label the new inputs will be classified based on prior training data.

Motivation

Overview

Course
Logistics

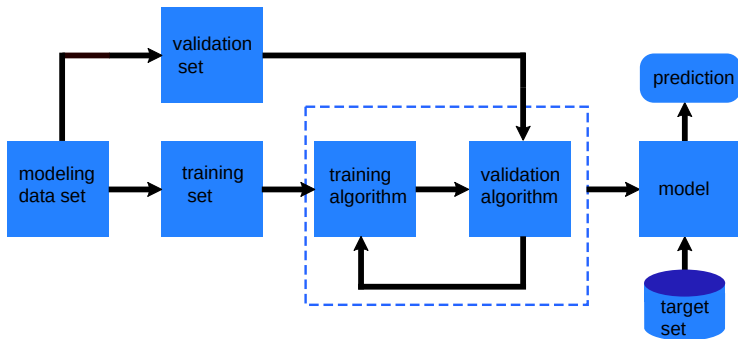
Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Supervised Machine Learning

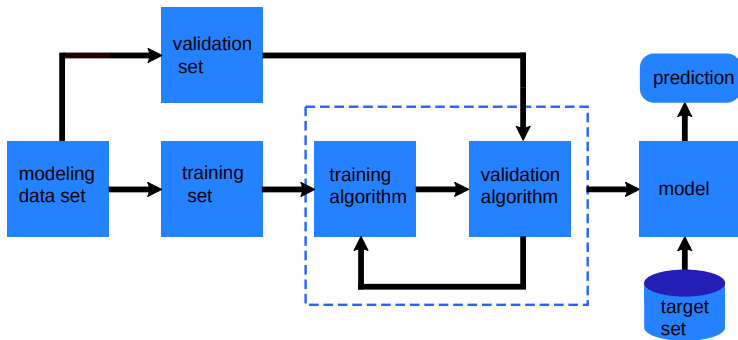


split modeling set into training set and validation set:

the validation set (also: test set) is used for the testing the model after the model has been trained on the training set - it is extremely important to test the model on data not being part of the training set

A **fundamental assumption** of supervised machine learning is that the distribution of training examples is identical to the distribution of validation examples and future unseen examples (the target set).

Supervised Machine Learning



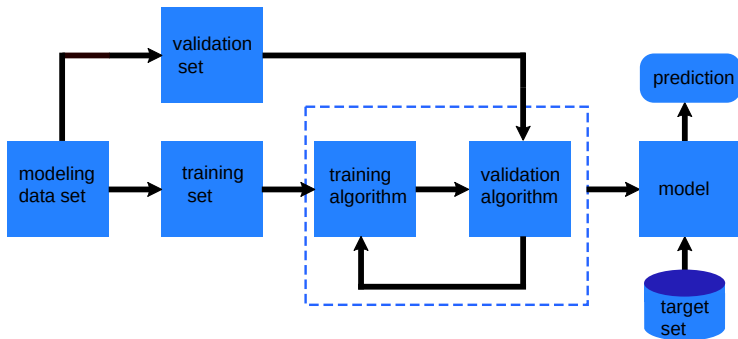
Training:

given a training set of labeled examples $\{(x_1, y_1), \dots, (x_n, y_n)\}$, estimate the prediction function f and parameters θ which minimizes the prediction error on the training set

Validation:

apply f to validation set x , output predicted value $y = f(x)$
from this we generate performance measures, also called accuracy measures

Supervised Machine Learning



Application:

Run the model on the target set.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

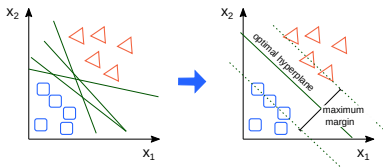
Software

Outlook

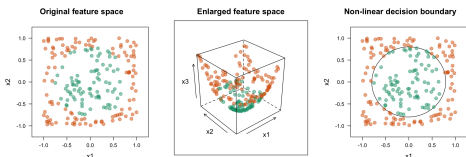
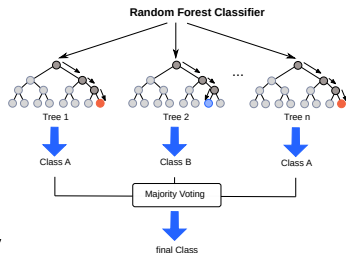
Supervised Machine Learning

state-of-the-art (before Deep Learning):

Support Vector Machines binary classification



Random Forest Classifiers multiclass classification



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Supervised Machine Learning

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

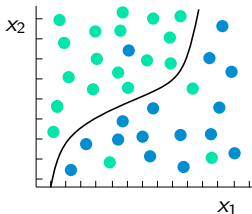
Machine
Learning -
Terminology

Software

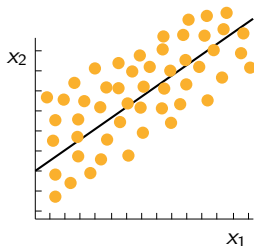
Outlook

two main areas of supervised machine learning:
classification problems and **regression** problems

mapping input value(s) to a
discrete value, the class
example: predicting whether
an object is a star or a galaxy



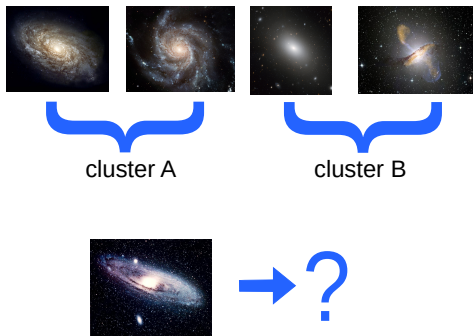
mapping input value(s) to
continuous data
example: predicting the
surface temperature of a star



Unsupervised Machine Learning

unlabeled data enable the unsupervised machine learning algorithm to **understand the data** and **find patterns in data themselves**

data is clustered, new data is assigned to clusters



applications: data exploration, data clustering, anomaly

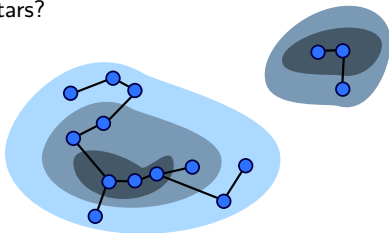
Unsupervised Machine Learning

three main areas of unsupervised machine learning:
clustering, **association** and **dimensionality reduction**

find hidden patterns in the data based on similarities or differences
example: are there subtypes within a given type of stars?

find the probability of co-occurrence of items in a collection
example: which stars likely host exoplanets?

reduce the dimensions of the data
example: feature retraction to reduce the number of random variables



Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

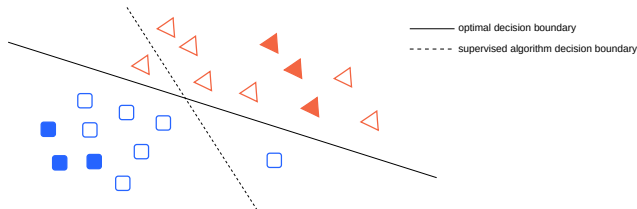
Software

Outlook

Semi-Supervised Learning

takes the **main advantages from both** supervised and unsupervised learning

it uses a **smaller labeled data set** to guide classification and performs unsupervised feature extraction from a **larger, unlabeled data set**



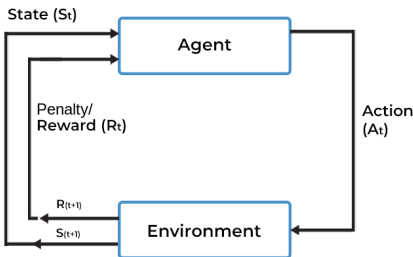
applications: solve problems when there is not enough labeled data present to train a model

Reinforcement Learning

an **agent** learns to behave in an environment by performing actions and adjusting its further course to feedback

data is not labeled, agent learns by experience: good actions are rewarded, bad actions result in a penalty

REINFORCEMENT LEARNING MODEL



applications: playing chess; optimizing astronomical survey

Types of Machine Learning - Overview

Machine Learning

Supervised Learning

Classification

table, time series classification, image classification

Regression

predictions, continuous variables

Unsupervised Learning

Clustering

anomaly detection, data exploration

Association

co-occurrence

Dimensionality Reduction

data visualization, feature extraction, structure discovery

Semi-supervised Learning

using small labeled data set and large unlabeled

Reinforcement Learning

robot navigation, survey strategy planning

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

The Role of Data in Training Process

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

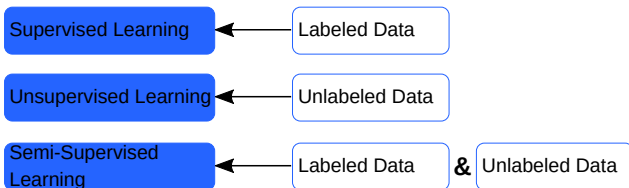
Machine
Learning -
Terminology

Software

Outlook

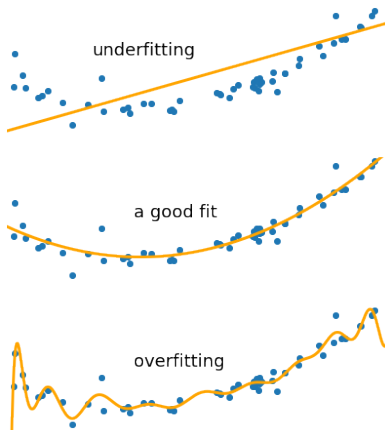
Supervised Learning learns from labeled training set data by iteratively making predictions on the data and adjusting for the correct answer. This makes supervised Learning models **more accurate** than unsupervised learning models.

Unsupervised Learning models work on their own to discover the inherent structure of unlabeled data. The unsupervised learning algorithm works with unlabeled data, in which the output is based solely on the collection of perceptions. This makes unsupervised methods **more flexible** to deal with new data.



Challenges and Limitations

In most scenarios, the cause of the poor performance of any machine learning algorithm is due to **underfitting or overfitting**.



Motivation

Overview

Course

Logistics

Machine
Learning in
Astronomy

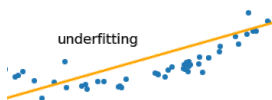
Machine
Learning -
Terminology

Software

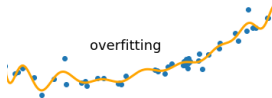
Outlook

Challenges and Limitations

In most scenarios, the cause of the poor performance of any machine learning algorithm is due to **underfitting or overfitting**.



Underfitting is a scenario where the machine learning model can neither learn the relationship between variables in the data nor predict a new data point correctly. In other words, the machine learning system hasn't found a correlation between data.



Overfitting occurs when the machine learning model learns from the training data a little too much, attempting to fit every point on the curve and, as a result, memorizes the data patterns. In other words, it narrowed its focus too much on the examples given, making it unable to see the bigger picture and fails to predict new data points.

Challenges and Limitations

Underfitting can occur when:

- The model was trained using the wrong features.
- The model is too simple and can't remember enough features.
- The target data is too varied or complex - the training set doesn't represent the target data's distribution realistically.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Challenges and Limitations

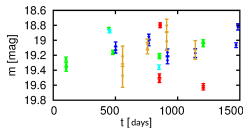
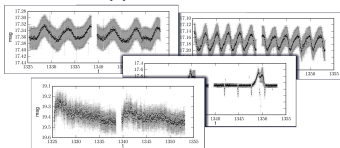
Underfitting can occur when:

- The model was trained using the wrong features.
- The model is too simple and can't remember enough features.
- The target data is too varied or complex - the training set doesn't represent the target data's distribution realistically.

Overfitting can occur when:

- The model was trained using the wrong parameters and over-observed the training data.
- The model complexity is too high for the presented data variability.
- The training data's labels are too restrictive.

example: Training on 'nice' (high cadence, long baseline, good S/N) light curves - applied to worse. Don't do that!



Key Takeaways: Machine Learning Basics

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

- Machine learning is a concept that allows computers to learn and improve from experience without being explicitly programmed.
- Machine learning works by the approach of *find the pattern, apply the pattern*.
- Machine Learning consists of Supervised, Unsupervised, Reinforcement, and Semi-Supervised Learning.
- Supervised learning is useful when dealing with purely labeled datasets for training and knowing how the output should look like.
- Unsupervised Learning is useful for finding the hidden patterns.
- A machine learning model is underfitted when it fails to capture the relationship between the input and output.
- If a machine learning model performs better on the training set than on the test set, then it is likely overfitting: it memorizes the data it was trained on without being able to generalize.
- Machine learning is part of many nowadays everyday applications such as Google Maps, Alexa, Youtube...
- It is increasingly important for astronomy for such as source classification, anomaly detection, survey strategy planning...

Software (Frameworks)

In this course, we will use plenty of "toy examples" to get a better understanding of how neural network algorithms work. We will write Python code for the inner behavior of neural networks.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Software (Frameworks)

In this course, we will use plenty of "toy examples" to get a better understanding of how neural network algorithms work. We will write Python code for the inner behavior of neural networks.

For any larger application, we will use software frameworks such as TensorFlow, astroNN or PyTorch.

For your **project** as part of this course, you can use whatever framework you want to use.

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Software (Frameworks)

TensorFlow

<https://www.tensorflow.org/>

TensorFlow is tested and supported on the following 64-bit systems:

Python >3.7

Ubuntu 16.04 or later

Windows 7 or later (with C++ redistributable)

macOS 10.12.6 (Sierra) or later (no GPU support)

The installation guide can be found here:

<https://www.tensorflow.org/install>

As behavior of TensorFlow, including compatibility with other libraries, can drastically change from version to version, I strongly advise to use **environments** in Python to keep your library versions stable.

Motivation

Overview

Course

Logistics

Machine

Learning in

Astronomy

Machine

Learning -

Terminology

Software

Outlook

Software (Frameworks)

TensorFlow

<https://www.tensorflow.org/>

TensorFlow can also be used on Google Colab (for testing purposes): Run the TensorFlow tutorials directly, without any installation, in the browser with Colaboratory, a Google research project created to help disseminate machine learning education and research. It's a Jupyter notebook environment that requires no setup to use and runs entirely in the cloud

<https://www.tensorflow.org/tutorials>

The TensorFlow tutorials are written as Jupyter notebooks and run directly in Google Colab. Just click the Run in Google Colab button.

<https://www.tensorflow.org/tutorials/quickstart/beginner>

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

Software (Frameworks)

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook

I strongly recommend using a Python **virtual environment** (e.g. `venv`) when working with such frameworks:

Often, functionality of those frameworks can change drastically from version to version, as well as they usually have very specific dependencies.



You do not want to encounter your code not longer working after an update!

An Outlook: Neural Networks & Deep Learning

three **ingredients**:

- discriminative neural network models (supervised learning)
- large labeled datasets
- lots of computer power

in particular **useful** for:

working with data sets for which computers typically don't perform well and specific algorithms are hard to write

- images
- videos
- speech/ time series data



ideal for **mining large astronomical survey datasets**

Motivation

Overview

Course
Logistics

Machine
Learning in
Astronomy

Machine
Learning -
Terminology

Software

Outlook